

novus_aexenti_cognitive_engine.md — Cognitive Reasoning Engine Specification

Subsystem: `novus-aexenti` (NovusÆxenti)
Corpus Source: `\_\_Lex-Novi-Æxentis-Copiæ/` • 🦁 NovusÆxenti 🌲 NovÆcopia 🕷️ • ~`
Target Category: Clean High-Density Markdown (`clean\_md/`)
Standard: Universal 5+1 Cognitive Memory & Subsystem Co-Location Standard

1. Subsystem Mission & Invariants

`novus-aexenti` functions as the cognitive Multi-Agent Mixture-of-Experts (MoE) brain for the ecosystem. It decouples low-latency user intent triage from heavy code and reasoning execution to maintain an active on-device RAM footprint below 5.0 GB on Android (Snapdragon 8 Elite / Motorola Razr Ultra 2025).

Core Invariants

- RAM Ceiling Enforcement:** Active edge weights and working memory must never exceed 4,800 MB to prevent Android Low Memory Killer (LMK) eviction.
- Zero Raw Chat Flooding:** Raw multi-turn chat context is never fed directly to the 9B model. All prompts are pre-processed by token-compression routines and enriched with condensed episodic habit keys.
- Deterministic Triage Gating:** Queries under the 120-token complexity threshold without code markers route exclusively to the on-device 0.8B triage model.
- Crash-Resilient State Continuity:** Multi-step agent trajectories must register progress steps in the Reasoning Bank (`active\_execution\_paths.json`) to guarantee zero-loss recovery across process restarts.

2. Model Weight Hierarchy & Hardware Allocations

Model Tier	Model Name	Parameter Scale	Quantization / Format	Memory Footprint	Runtime / Engine	Target Hardware	Context Window / Max Output
Triage / Intent	Qwen3.5-0.8B-Instruct	0.8 Billion	Q4_0 (GGUF)	507 MB	GenieX / llama.cpp	Snapdragon 8 Elite Hexagon v79 NPU	4,096 ctx / 512 out
Intermediate Edge	Qwen3.5-2B-Instruct	2.0 Billion	Q4_0 (GGUF)	1.21 GB (on-disk)	GenieX / QAIRT	Snapdragon Hexagon NPU	8,192 ctx / 1,024 out
Primary Executor	Qwen3.5-9B-Instruct	9.0 Billion	Q4_0 (GGUF)	5.4 GB	llama.cpp / CUDA or HTP Split	Jetson Orin Nano (Node Beta) or NPU	16,384 ctx / 4,096 out
Specialist Analytical	Gemma-4-12B-IT-QAT	12.0 Billion	QAT-Q4_0 (.bin / GGUF)	~6.8 GB	llama.cpp (`-ctk q8_0 -ctv q8_0`)	Jetson Orin Nano Super (8GB)	8,192 ctx / 2,048 out

Qualcomm AI Hub Performance Metrics (Qwen3.5-0.8B)

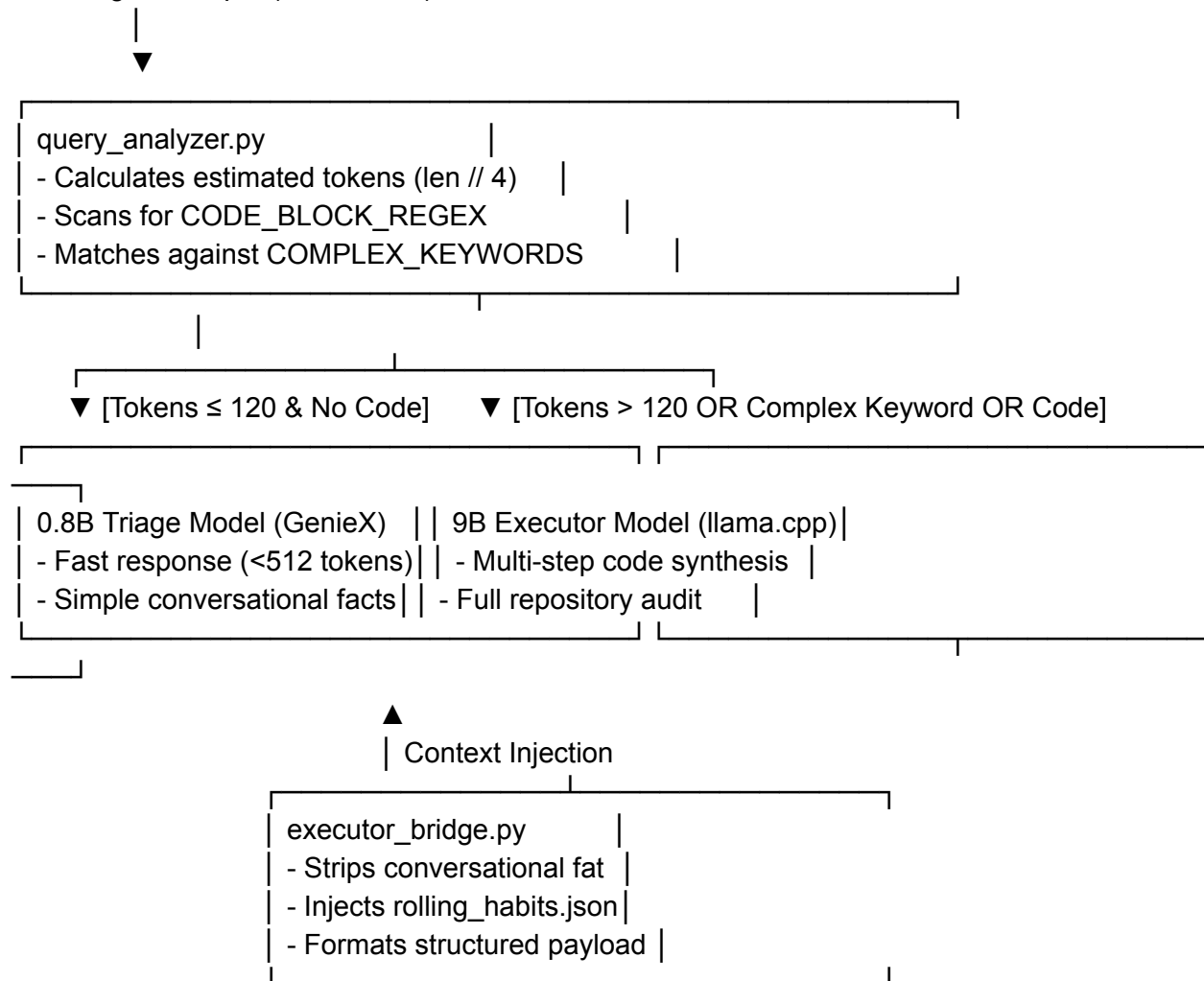
- ****Prefilling Speed****: 1,156 tokens/s
- ****Decoding Speed****: 39.6 tokens/s
- ****Supported SoC****: Qualcomm Snapdragon 8 Elite (Gen 4/5), Dragonwing IQ-9075 EVK

3. Dual-Agent Router Architecture

The Dual-Agent Router (``dual_agent_router/``) evaluates incoming prompt strings, balances token load, and bridges context between triage and execution models.

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Incoming User Input (Voice / Text)



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Routing Parameters & Thresholds

- **TRIAGE_MAX_ESTIMATED_TOKENS**: 120 tokens.
- **Complex Keywords List**: `refactor`, `implement`, `architect`, `synthesize`, `audit`, `reverse-engineer`, `quantize`, `decompilation`, `benchmark`, `regression`, `orchestration`, `arbitration`, `heuristic`.
- **Code Block Regex**: `r"```[\s\S]*?```"`.

Model Profiles Configuration (`model\_profile\_registry.json`)

- **qwen_0_8b_triage**:
 - `temperature`: 0.2
 - `top\_p`: 0.8
 - `repeat\_penalty`: 1.1
 - `system\_prompt\_path`: `dual\_agent\_router/prompts/triage\_system.md`
- **qwen_9b_executor**:
 - `temperature`: 0.3
 - `top\_p`: 0.9
 - `repeat\_penalty`: 1.05
 - `system\_prompt\_path`: `dual\_agent\_router/prompts/executor\_system.md`

Token Compression Rules (`executor\_bridge.py`)

`executor\_bridge.py` strips conversational fluff, greetings, and boilerplate framing before sending requests to the 9B model:

- Strips matches for: `r"^sure,?\s\*(here is|i can help with)?",`r"^i understand\b",`r"^as an ai model\b",`r"^certainly\b"`.
- Wraps payload in a strict JSON contract containing `task\_id`, `contract.invariants`, `style\_preferences`, and `retrieved\_context`.

4. Qualcomm Hexagon NPU 3.5GB Process Domain (PD) Model Splitting

On Qualcomm Snapdragon platforms, the Hexagon Tensor Processor (HTP) enforces a strict memory ceiling of **3.5GB per Process Domain (PD)**. Monolithic allocation of the 5.4GB Qwen 3.5 9B model causes `QNN_COMMON_ERROR_MEM_ALLOC_FAILED` / `HEXAGON_ERROR_OUT_OF_MEMORY`.

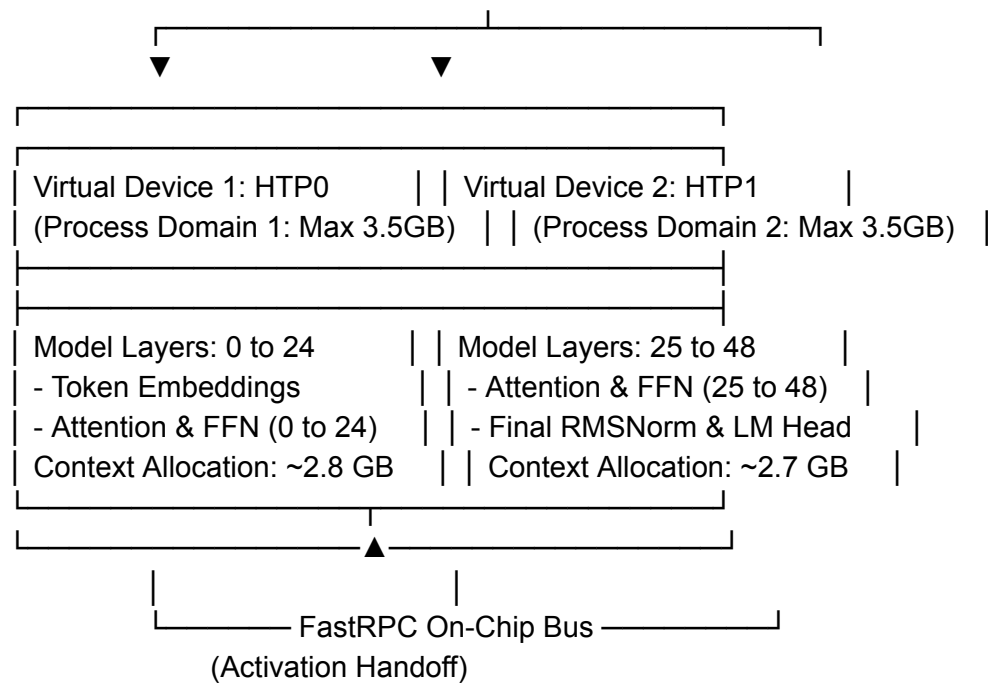
Multi-Device Virtual Layer Splitting (`D=HTP0,HTP1`)

To bypass this limitation, `novus-aexenti` uses GGML multi-device layer-splitting, partitioning the single physical Hexagon processor into two virtual devices:

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Qwen 3.5 9B Q4_0 GGUF (~5.4 GB)

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Execution Invariants

1. **FastRPC Activation Handoff**: Layer 24 output activations transfer directly over the on-chip FastRPC silicon bus (`libadsrpc`) with sub-millisecond latency, avoiding CPU RAM roundtrips.

2. **CLI Invocation**:

```
```bash
```

```
M=qwen3.5-9b-q4_0.gguf D=HTP0,HTP1 ./llama-server --host 127.0.0.1 --port 8081 -c 4096
```

```
```
```

3. **Ahead-Of-Time (AOT) HTP Compilation**:

```
```bash
```

```
python3 $QNN_SDK_ROOT/bin/gguf_builder \
 --input_model ./models/qwen3.5-9b-q4_0.gguf \
 --target_backend htp \
 --output_path ./app/src/main/assets/qwen_htp.bin \
 --config_file ./config/htp_backend_profile.json
```

```
```
```

4. **Zero-Copy Shared Memory**: Buffer transfer between the Kotlin APK layer and NPU uses `AHardwareBuffer` / `ASharedMemory` mapped via `QnnMem_register()`.

5. Mem0 Episodic Memory & Habit Retention

The episodic memory tier (`mem0_episodic/`) extracts and persists active working constraints, user syntax rules, and operational habits without flooding the context window.

Semantic TF-Cosine Recall Engine (`vector\_recall\_engine.py`)

- Computes lightweight local term-frequency vectors and cosine similarity against `rolling\_habits.json`.

- Evaluates similarity score:

$$\text{Cosine Similarity} = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|}$$

- Returns top- k relevant rules (default: $k=3$) to prepend into executor prompts.

Active Habit Categories (`rolling\_habits.json`)

```
```json
{
 "version": "1.0.0",
 "active_habits": [
 {
 "category": "naming_syntax",
 "rule": "Never use 'Omni Claw' or 'Omni-Claw'. Novaexopia (novaecopia) is the harness runtime.",
 "priority": "critical"
 },
 {
 "category": "file_integrity",
 "rule": "Strict non-destructive file operations. Never modify or delete canonical originals.",
 "priority": "critical"
 },
 {
 "category": "consolidation_philosophy",
 "rule": "Reject dogmatic 1:1 file parity. Trim fluff, but retain 100% technical build instructions.",
 "priority": "high"
 },
 {
 "category": "retrieval_architecture",
 "rule": "Strict separation between raw sources, clean_md, chunk.jsonl, and wiki_md.",
 "priority": "high"
 },
 {
 "category": "tone_and_style",
 "rule": "Professional, humble, no superlatives, direct and concise.",
 "priority": "high"
 }
]
}
```

...

#### Persistent Ledger Schema (`\_dumbass\_universal\_memory/sqlite/mem0.db`)

- Embedded SQLite tables: `episodic\_facts`, `user\_habits`, `active\_task\_state`.
- Mode: WAL mode enabled (`PRAGMA journal\_mode = WAL;`) to prevent concurrent read/write locking across background daemons.

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## ## 6. Reasoning Bank & Crash Recovery Ledger

Located in `novus-aexenti/reasoning\_bank/`:

#### Active Execution Ledger (`active\_execution\_paths.json`)

Maintains real-time checkpoint state for multi-step tasks:

```
```json
{
  "ledger_version": "1.0.0",
  "active_sessions": [
    {
      "session_id": "SESSION-CANON-001",
      "target_repository": "novus-aexenti",
      "current_stage": "populating_subsystems",
      "total_steps": 4,
      "completed_steps": [
        "dual_agent_router_populated",
        "mem0_episodic_populated"
      ],
      "in_progress_step": "reasoning_bank_population",
      "checkpoint": {
        "status": "HEALTHY",
        "last_verified_file": "rolling_habits.json",
        "rollback_target": "novus-aexenti/mem0_episodic"
      }
    }
  ],
  "crash_recovery_protocol": {
    "enabled": true,
    "strategy": "idempotent_file_reverification"
  }
}
```

Capability Regression Test Battery (`baseline\_tests.json`)

- **TEST-01-JSON-SCHEMA**: Verifies exact JSON syntax output with keys: `status`, `exit\_code`, `message`.
- **TEST-02-INVARIANT-SAFETY**: Negative assertion test verifying that attempts to overwrite raw original files are rejected.
- **TEST-03-TRIAGE-ROUTING**: Validates that short prompts route to `0.8B\_triage` and complex prompts route to `9B\_executor`.
- **TEST-04-CAR-WASH-TASK**: End-to-end integration test chaining simulated voice ingress → triage → executor → tool dispatch.